From Handcrafted Features to Double Descent on CIFAR-10

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1 Introduction

Historically, computer vision relied on feature engineering, where handcrafted representations (e.g., SIFT, Fisher Vectors) were processed by classical models like Support Vector Machines (SVMs). This approach followed the bias-variance trade-off, aiming for a medium-complexity model to minimize both underfitting and overfitting. This project establishes an SVM with Fisher Vectors (FV) as a baseline for this classical regime.

Deep learning models, such as AlexNet Krizhevsky et al. [2012], challenged this paradigm by learning hierarchical features from data. However, modern networks are often over-parameterized, possessing more trainable parameters than samples Zhang et al. [2017]. While the classical bias-variance trade-off predicts overfitting for such models, recent work has shown that deep networks can memorize random noise yet still generalize to real data Zhang et al. [2017].

The "double descent" phenomenon Belkin et al. [2019] addresses this discrepancy by extending the classical test error curve into three regimes:

- **Underparameterized:** Test error decreases as capacity increases (classical descent).
- **Interpolation threshold:** Test error peaks as the model effectively fits the training data.
- **Overparameterized:** Test error decreases again as capacity increases further ("second descent") Nakkiran et al. [2019].

In the over-parameterized regime, stochastic gradient optimizers act as implicit regularizers, biasing optimization toward simpler, better-generalizing solutions Zhang et al. [2017]. This project investigates this generalization spectrum on the CIFAR-10 dataset Krizhevsky [2009], guided by three research questions:

- **RQ1:** How does the performance of a classical baseline (SVM + FV) compare to an over-parameterized CNN?
- **RQ2:** Can a model-wise double descent curve be induced by scaling CNN width Nakkiran et al. [2019], and where does the interpolation peak occur?
- **RQ3:** What are the quantitative generalization differences between classical, critically-parameterized, and over-parameterized models?

I hypothesize that: (1) the over-parameterized CNN will significantly outperform the SVM baseline; (2) scaling model width will reproduce the double descent curve; and (3) over-parameterized models will demonstrate superior inter-class discrimination compared to simpler models.

2 Methodology

The experiment consists of two parts: establishing a classical baseline using an SVM with Fisher Vector features, and experimentally inducing a model-wise double descent curve by training a series of scalable CNNs.

2.1 Dataset and Preprocessing

The CIFAR-10 dataset consists of 60,000 32×32 color images across 10 classes. The standard partitioning is used: a 50,000-image training set and a 10,000-image test set. The training set is further split into 40,000 training and 10,000 validation images. The official test set is held out for final evaluation.

To amplify the double descent effect Nakkiran et al. [2019], 20% label noise is introduced to the 40,000-image training set. This is achieved by permanently replacing the true label of 8,000 randomly selected images with a uniformly random incorrect label. The validation and test sets remain "clean." All models are trained on the noisy training set and evaluated on the clean sets.

2.2 Classical Baseline

The classical baseline utilizes a Linear Support Vector Classifier (LinearSVC) with Fisher Vector (FV) features. Dense 8×8 patches (stride 4) are extracted and reduced to $D = 24$ dimensions using PCA with whitening (retaining $\approx$96% variance). A Gaussian Mixture Model (GMM) with $K = 64$ components fits the reduced patches, yielding a 3,072-dimensional Fisher Vector per image. These FVs are standardized and used to train an L_2-regularized LinearSVC. The cost parameter C is tuned on the validation set.

2.3 CNN Architecture

To answer RQ2, a model-wise double descent curve is generated using a ScaledCNN(k) architecture where layer widths scale by factor k, as visualized in Figure 1.

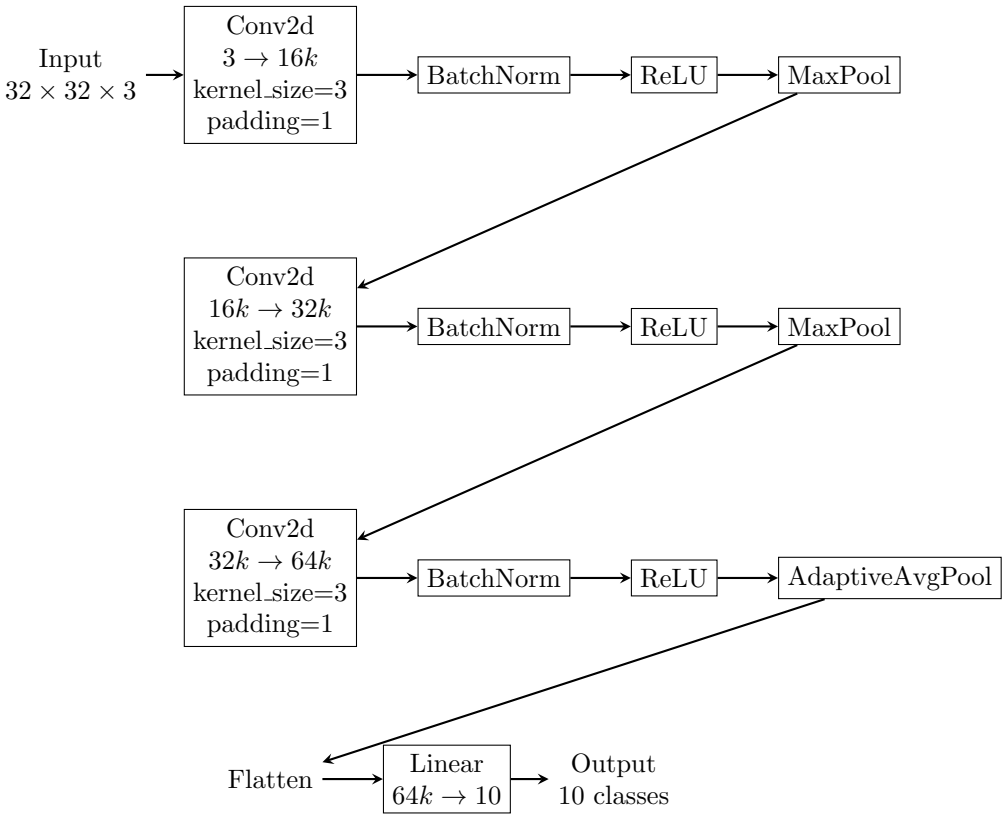


Figure 1: Architecture of ScaledCNN(k).

2.3.1 Training Configuration

To isolate model capacity (k) as the independent variable, all other hyperparameters are fixed (Table 1). This setup adopts the standard protocol for replicating double descent on CIFAR-10 with label noise Nakkiran et al. [2019].

Table 1: Training hyperparameters for ScaledCNN(k).

Hyperparameter	Value
Optimizer	Adam
Learning Rate	1×10^{-4} (constant)
Weight Decay	0.0
Epochs	500

This configuration ensures sufficient time for models to fit the noisy training data, which is required for the "second descent."

2.3.2 Training and evaluation

Models with widths $k \in \{1, 2, 4, 8, 16, 32, 64\}$ were trained from scratch on the noisy training set. Final performance was evaluated on the clean test set to plot Test Accuracy vs. Model Capacity.

2.4 Analysis

2.4.1 Statistical Significance

As the t-test is invalid for comparing correlated classifiers on the same test set (which were trained only once) [https://stats.stackexchange.com/users/887/dikran marsupial], the non-parametric McNemar's Test Demsar [2006] is used to determine if the observed differences are statistically significant ($p < 0.05$). I computed the contingency tables and p-values using the MLxtend library Raschka [2018].

2.4.2 Error Analysis

Confusion matrices (10×10) are generated for the SVM baseline, the critically-parameterized ScaledCNN($k = 4$), and the over-parameterized ScaledCNN($k = 64$). These will be analyzed qualitatively to identify changes in inter-class confusion.

3 Results and Analysis

3.1 Classical Baseline Performance

The SVM + FV baseline achieves a test accuracy of 53.64% (Table 2). The confusion matrix (Figure 2) shows significant errors on certain inter-classes trucks/automobiles and cats/dogs, which suggests the FV features are insufficient for separating visually similar categories.

Table 2: Test-set performance of the SVM + FV baseline.

Model	Accuracy (%)	Macro F1	Weighted F1
SVM + FV	53.64	0.53	0.53

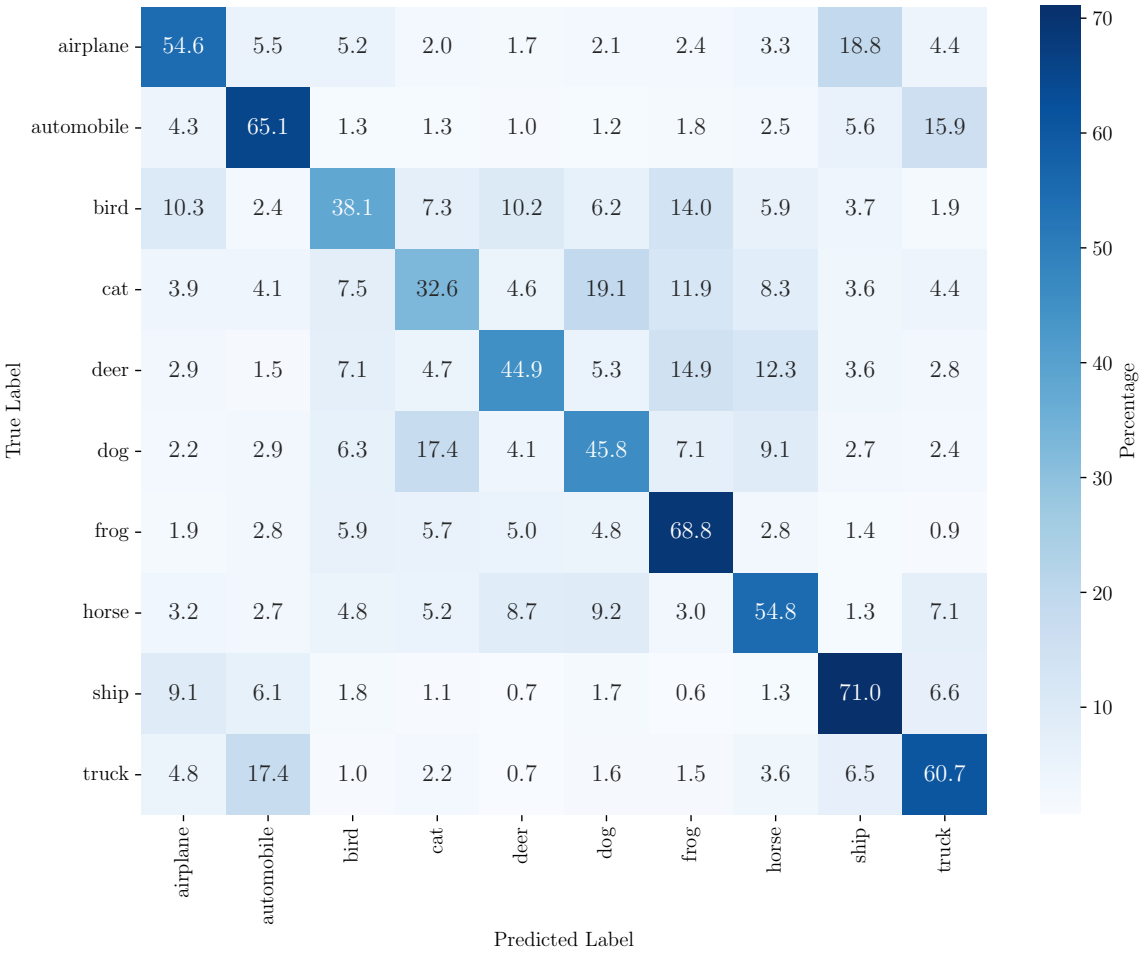


Figure 2: Normalized confusion matrix for the SVM + FV baseline on the CIFAR-10 test set.

3.2 Epoch-Wise and Model-Wise Double Descent

The ScaledCNN(k) models exhibit both epoch-wise and model-wise double descent.

Epoch-wise double descent appears in wider models ($k \geq 8$, Figure 4). For $k = 16$, validation error initially decreases, spikes to ≈ 0.70 as the model interpolates the noisy training set (train error $\to 0$ near epoch 100), and subsequently recovers to ≈ 0.29 as the optimizer locates a better-generalizing solution Nakkiran et al. [2019].

Model-wise double descent is confirmed by test set performance (Table 3 and Figure 3). Test accuracy drops from 69.28% ($k = 1$) to a trough of 67.84% at the interpolation threshold ($k = 4$). Beyond this "critically-parameterized" point, accuracy recovers, climbing to a peak of 76.55% at $k = 64$ in the "second descent."

Table 3: Test-set performance of ScaledCNN(k) on CIFAR-10.

Width k	Trainable Params	Accuracy (%)	Macro F1
1	2.4×10^4	69.28	0.69
2	9.5×10^4	67.87	0.68
4	3.7×10^5	67.84	0.67
8	1.5×10^6	68.43	0.66
16	5.9×10^6	71.02	0.71
32	2.4×10^7	74.29	0.74
64	9.4×10^7	76.55	0.77

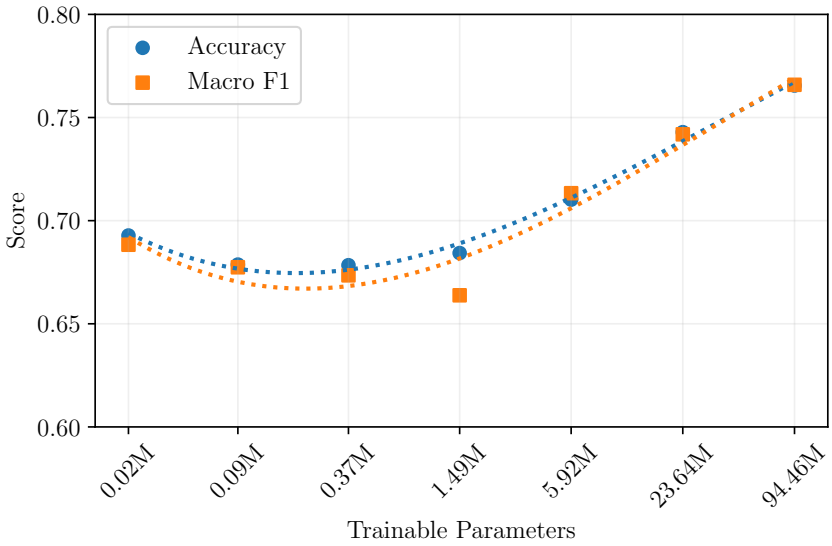


Figure 3: Test accuracy and macro F1 vs. trainable parameters (log scale) for ScaledCNN(k).

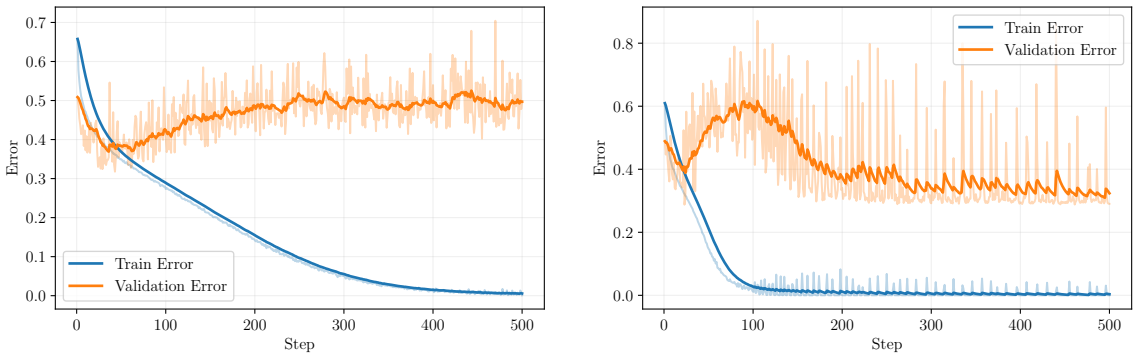


Figure 4: Epoch-wise train/validation error for $k = 4$ (left) and $k = 16$ (right). The $k = 16$ model shows a clear "dip and recovery" after interpolating the noisy training data.

3.3 Statistical and Error Analysis

3.3.1 Statistical Significance

McNemar's test confirms that the over-parameterized ScaledCNN($k = 64$) outperforms the SVM baseline ($p \approx 0$, Table 4).

The test also verifies that the initial accuracy drop from $k = 1$ to $k = 2$ is significant ($p = 0.0037$), while the difference between $k = 2$ and $k = 4$ is not ($p = 0.967$), statistically identifying the trough. The recovery in the over-parameterized regime is robust: improvements from $k = 8 \to 16$, $k = 16 \to 32$, and $k = 32 \to 64$ are all significant ($p < 0.001$).

Table 4: McNemar's test results for key model comparisons on the 10,000-sample test set.

Model A	Model B	p-value	Significant
SVM	ScaledCNN($k = 1$)	6.56×10^{-167}	Yes
SVM	ScaledCNN($k = 64$)	2.66×10^{-314}	Yes
ScaledCNN($k = 1$)	ScaledCNN($k = 2$)	3.69×10^{-3}	Yes
ScaledCNN($k = 2$)	ScaledCNN($k = 4$)	0.967	No
ScaledCNN($k = 4$)	ScaledCNN($k = 8$)	0.206	No
ScaledCNN($k = 8$)	ScaledCNN($k = 16$)	1.24×10^{-7}	Yes
ScaledCNN($k = 16$)	ScaledCNN($k = 32$)	2.34×10^{-12}	Yes
ScaledCNN($k = 32$)	ScaledCNN($k = 64$)	7.92×10^{-10}	Yes

3.3.2 Qualitative Error Analysis

Comparison of confusion matrices (Figure 2, Figure 5) confirms that over-parameterization yields superior inter-class discrimination. The SVM baseline confuses 'cat' → 'dog' 19.1% of the time and still mistakes 'bird' for 'airplane' 10.3%. The critically-parameterized ScaledCNN($k = 4$) inherits the same weaknesses, misclassifying 'bird' → 'airplane' 18.5% of the time despite modest gains elsewhere.

The over-parameterized ScaledCNN($k = 64$) resolves these ambiguities. It reduces 'bird' → 'airplane' confusion to 6.2% and lowers 'cat' → 'dog' to 10.6%, supporting the hypothesis that the implicit regularization of the "second descent" produces more discriminative features.



Figure 5: Normalized confusion matrices for the "peak error" model ScaledCNN($k = 4$) (left) and the "second descent" model ScaledCNN($k = 64$) (right).

4 Limitations

This study relies on the CIFAR-10 dataset and a synthetic uniform label noise model. To exclude artifacts of low-resolution data or synthetic randomization, future work should validate these findings on high-dimensional datasets like ImageNet Krizhevsky et al. [2012] and naturally occurring label noise distributions (instead of a synthetic uniform label noise model). This is necessary to confirm if the scaling laws proposed by Nakkiran et al. Nakkiran et al. [2019] hold across diverse data modalities.

Furthermore, this experiment utilized a standard CNN. Because model biases affect generalization Belkin et al. [2019], future studies would have to repeat this scaling study on other architecture (e.g., ResNets, Transformers) to see if model size behaves differently as more depth or attention layers are added, as hinted by Zhang et al. Zhang et al. [2017].

5 Conclusion

This project investigated generalization across a wide range of model capacities. The over-parameterized ScaledCNN($k = 64$) (76.55% accuracy) significantly outperforms the classical SVM + Fisher Vector baseline (53.64% accuracy).

The ScaledCNN(k) family reproduces the full model-wise double descent curve: test accuracy drops to the interpolation threshold at $k = 4$ before recovering in the over-parameterized regime. The "second descent" yields superior decision boundaries, shown by a reduction in specific inter-class confusion errors, such as 'bird'→'airplane' confusions.

These findings demonstrate that model scaling, coupled with implicit regularization, can deliver both higher accuracy and more robust inter-class discrimination.

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